PRD: Customer Application Assistant (Working Title)

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This draft assumes a regulated, form-based application journey. Product name, customer base and regulatory scope are not yet confirmed. All quantitative figures are hypothetical and must be replaced with validated baselines.

1. About

Customers in multi-step application journeys struggle to know where they stand, what to do next, and why information is requested. They rely on static help pages, phone queues, and repeated retelling of their situation to agents. This PRD defines a contextual assistant that shows verified application progress, explains the current screen or requirement, recovers from common errors, hands off to humans with full context, and explains data requests in plain language, without pushing sales.

TL;DR: Make application status and the next required action visible and trustworthy, so customers complete journeys without contacting support.

2. Market Insights

- **Competitor analysis:** [To validate] Survey 3–5 incumbents for status trackers, in-flow help, and AI assistants. Record whether each uses verified system-of-record data or generative answers only.
- **Market analysis:** [To validate] Assess saturation. Early signal suggests many providers offer chatbots, but few combine verified status with a constrained, non-sales assistant.
- **Technology analysis:** Most rely on retrieval-augmented chat over help content. Few ground answers in live application data. Differentiation lies in grounding, error taxonomy, and handoff context, not in the model itself.
- **Customer segments:**
  - *Primary:* New applicants mid-journey, with one or more abandoned or stalled applications.
  - *Secondary:* Returning applicants with outstanding documents.

- *Tertiary:* Customers who need human help but lack an account number.

User persona: "Priya, the Stalled Applicant"

- Age 28–45, mobile-first, limited time, moderate digital literacy.
- **Goal:** Finish the application in one session without calling anyone.
- **Frustrations:** Unclear status, unexplained document requests, errors she cannot interpret, and having to repeat herself to agents.
- **Trust threshold:** Will not act on information she cannot verify.

3. The Problem

Use cases

1. Check where my application stands and what is still outstanding.
2. Understand what a field or document request means and why it is needed.
3. Recover from a specific error without starting over.
4. Reach a human when the assistant cannot resolve my issue, without repeating myself.
5. Understand how my data is used and why a request is being made.

Pain points

- Status is ambiguous or delayed. Customers cannot distinguish "under review" from "blocked."
- Requirement explanations are generic or buried in help pages.
- Common errors (e.g., mismatched name formats, expired upload links) generate repeat contacts.
- Handoffs lose context. Customers must re-authenticate and retell their case.
- Customers distrust data requests, especially when the purpose is unclear.
- Assistants that mix sales messaging with service erode trust.

Problem statement

Priya spends too long guessing what is blocking her application and what to do next, and she often abandons it or calls support, because the journey does not show verified progress or plain-language explanations.

Hypotheses and mission statement

- **H1:** If we show verified status and the next required action, self-serve completion will rise and status-related contacts will fall.
- **H2:** If we explain requirements in context, field-level drop-off will fall.
- **H3:** If handoffs carry structured context, average handling time (AHT) and repeat-explanation rates will fall.

Mission: By bringing a verified, explainable assistant to life, we will make finishing an application faster, clearer, and more trustworthy for the customer.

4. The Solution

Ideation: prioritised features

Priority	Feature	Minimum useful version
1	Reliable status and next step	Verified progress, outstanding actions, known blockers
2	Contextual conversational help	Explain current screen or requirement without forcing chat
3	Failure recovery	Recognise a small set of frequent errors; offer supported actions
4	Human handoff with context	Escalate without account number or full retelling
5	Trust and data explanations	Explain requests, flag uncertainty, separate assistance from sales

Leveraging AI

How can we use AI to help users?

- **Interpret context:** Map the current screen and application state to plain-language explanations.
- **Classify errors:** Match free-text or system error codes to a small, governed taxonomy.
- **Summarise for handoff:** Produce a structured case summary so customers never retell their situation.
- **Explain requirements:** Rephrase approved policy text at an appropriate reading level.

Why AI is essential: Requirement and error explanations vary by product, journey stage, and customer state. Static rules cannot cover this combinatorial space at reasonable maintenance cost. AI must be **grounded**: status claims come only from verified system data via tools, and policy explanations come only from approved content. The model never invents status or eligibility.

Feature prioritisation (RICE)

Reach is the percentage of the target persona affected. Impact, Confidence, and Effort are scored 1–10. Score = (R × I × C) / E.

Feature	Reach	Impact	Confidence	Effort	Score
1. Reliable status and next step	90%	9	8	6	**1,080**
2. Contextual conversational help	60%	7	6	5	**504**
3. Failure recovery	40%	8	7	4	**560**
4. Human handoff with context	35%	8	8	6	**373**
5. Trust and data explanations	70%	6	7	3	**980**

Analysis: RICE ranks Feature 5 second and Feature 3 above Feature 2, because both are cheap to build. The strategic order keeps Feature 2 above Feature 3, since contextual help affects every stage of the journey. **Recommendation:** Keep Features 1 and 5 in the first release. Sequence Features 2 and 3 as planned. Revisit after pilot data.

AI MVP

- **Model approach:** A hosted LLM orchestrated through a tool-calling layer. Tools include `get_application_status`, `get_outstanding_actions`, `get_screen_context`, and `get_approved_content`.
- **Grounding:** Status and blocker data come from the system of record. Policy text comes from a governed, versioned knowledge base. Retrieval is limited to approved content.
- **Error taxonomy:** A rules-first classifier covers the top 10–15 error codes. The LLM handles residual free text and maps it to taxonomy entries.
- **Guardrails:** Output filters block sales language, PII echo, and unsupported status claims. When confidence is low or data is unavailable, the assistant states uncertainty and offers handoff.
- **Scope exclusions:** No eligibility decisions, no pricing or product recommendations, no autonomous actions on the customer's behalf in MVP.

Roadmap

> *[Add image from Canva template]*

Phase	Timing (hypothetical)	Scope
Phase 0: Foundations	Nov 2026	Status API integration, approved-content store, telemetry
Phase 1: MVP	Jan 2027	Features 1 and 5; error recovery for top 5 codes
Phase 2: Expansion	Mar 2027	Feature 2 (contextual help); Feature 3 full taxonomy
Phase 3: Handoff	May 2027	Feature 4 with structured case summaries

Technical architecture (high level)

- **Client:** Web and mobile application widget. Sends the current screen identifier and an authenticated session token. Never sends raw PII in the prompt.
- **Assistant service:** Orchestrates LLM calls, enforces guardrails, and logs telemetry.
- **Data services:** Application status API (read-only), outstanding-actions service, content store (approved policy text with version IDs).
- **LLM provider:** Hosted inference, with data-processing terms reviewed by Legal.
- **Handoff:** Contact-centre platform receives a structured case summary, with conversation context linked to the case.
- **Telemetry pipeline:** Event stream to the analytics warehouse, with PII tokenised.
- **Data flow:** Client → Assistant service → (Status API, Content store) → LLM → Assistant service → Client. Handoff: Assistant service → Contact-centre platform.

Software engineers and data scientists will produce the detailed design document.

Assumptions and constraints

- **Assumptions:**
  - Application status data is available with 15-minute latency or less for at least 95% of records.

  - Approved policy content exists or can be produced by Compliance within 6 weeks.
  - Customers have authenticated sessions for status features.
- **Constraints:**
  - Assistance must remain separate from sales. Sales offers cannot appear in assistant responses.
  - Responses must be consistent with regulatory disclosure requirements.
  - Latency budget: p95 under 2.5 seconds per response.
  - Languages: English at launch; additional languages scoped in Phase 2.

Risks

Risk	Likelihood	Impact	Mitigation
Assistant states incorrect status	Medium	High	Grounding via tools only; reconciliation checks; visible data timestamp
Stale status displayed as current	Medium	High	Show "as of" timestamp; flag data older than threshold
Customer shares sensitive data in chat	High	High	PII masking; prompt-level warnings; no storage of raw IDs
Regulatory breach from policy wording	Low	High	Approved-content-only retrieval; Compliance review gate
Over-reliance; customers stop calling when they should	Medium	Medium	Always-visible handoff path; monitor complaint rates
Handoff summary contains errors	Medium	Medium	Structured fields; agent confirmation before action

5. Requirements

User stories

- As an applicant, I want to see my verified progress and next required action so I can finish without calling.

- As an applicant, I want an explanation of a requirement on my current screen so I understand what to provide.
- As an applicant, I want a clear recovery path when I hit a known error so I do not restart.
- As an applicant, I want to reach a human without re-entering everything so I save time.
- As an applicant, I want to know why information is requested and how it is used so I can decide whether to proceed.

Functional requirements

F1. Reliable status and next step

- Display verified stage, outstanding actions, and known blockers from the status API.
- Show data timestamp; warn when data is older than 15 minutes.
- If data is unavailable, state that clearly and offer handoff. Never infer status.
- *Data requirements:* `status_panel_viewed`, `next_action_displayed` (action_type), `blocker_displayed` (blocker_code), `status_data_age_seconds`, `status_unavailable_shown`, `next_action_completed` (within 24h).

F2. Contextual conversational help

- Assistant reads the current screen identifier and offers an explanation without requiring a chat opening.
- Explanations draw only from approved content and cite the version.
- *Data requirements:* `help_prompt_shown` (screen_id), `help_opened`, `help_helpful_rating`, `help_content_version`, `field_drop_off` (screen_id, field_id), `help_followup_question_category`.

F3. Failure recovery

- Classify errors against the taxonomy (top 10–15 codes at launch).
- Offer supported actions only (retry, re-upload, edit field, contact handoff). No workarounds that bypass controls.
- Unrecognised errors go to handoff with error details.
- *Data requirements:* `error_shown` (error_code, screen_id), `error_classified` (taxonomy_id, confidence_bucket), `recovery_action_offered`, `recovery_action_taken`, `error_resolved_within_session`, `error_repeat_within_7d`.

F4. Human handoff with context

- Available at all times. Does not require an account number. Offers optional verification via a secure path.
- Generates a structured summary: journey stage, screens visited, error codes, actions attempted, customer's stated goal. Customer reviews and confirms before submission.
- Customer can decline the summary.
- *Data requirements:* `handoff_offered`, `handoff_initiated` (reason_code), `handoff_summary_confirmed`, `handoff_summary_edited`, `handoff_queue_wait_seconds`, `agent_reexplanation_flag` (agent-reported), `handoff_csat`.

F5. Trust and data explanations

- Explain each information request: what, why, legal basis (where applicable), and what happens if not provided.
- Flag uncertainty explicitly ("I cannot confirm this from your records").
- No sales content. Recommendations are blocked by output filter.
- *Data requirements:* `data_request_explained` (field_id, explanation_version), `data_explanation_expanded`, `uncertainty_flag_shown`, `sales_content_blocked` (filter_count), `trust_survey_score`.

Non-functional requirements

- **Security:** Authenticated-only status features. Encryption in transit and at rest. Role-based access for logs.
- **Privacy:** PII tokenised in telemetry. Chat logs retained per policy (e.g., 90 days), then deleted. Customers can request deletion.
- **Performance:** p95 response under 2.5 seconds; status panel under 1 second.
- **Scalability:** Support 10× current contact peak without degradation.
- **Availability:** 99.9% for status features; graceful degradation to static help when the assistant is down.
- **Accessibility:** WCAG 2.2 AA; screen-reader compatible; keyboard operable.
- **Localisation:** English at launch; string externalisation for future languages.
- **Observability:** Dashboards for latency, classification confidence, guardrail triggers, and handoff rate.

AI and data requirements

- **Data sources:** Application status records, outstanding-actions records, screen and field metadata, error code catalogue, approved policy content (versioned), handoff queue data.
- **Data types:** Structured application state; semi-structured policy text; masked conversation transcripts; telemetry events.
- **Training and evaluation data:** Anonymised historical support transcripts (for taxonomy and intent evaluation), with PII removed. Gold-standard evaluation set of about 500 scenarios reviewed by Support SMEs.
- **Management processes:** Content versioning and approval workflow; data lineage documentation; quarterly review of taxonomy coverage; deletion workflows aligned with retention policy.

6. Challenges

- **Will we have enough data?** Historical transcripts may suffice for taxonomy design, but not for a full evaluation set. *Plan:* Sample 6 months of contact data and label it with SMEs.
- **How do we acquire it?** Internal support logs (with privacy review), content from Compliance and Product, and synthetic scenarios for edge cases.
- **Do we have funds?** [TBD] Estimate LLM inference, engineering, Compliance review, and contact-centre integration. Secure budget before Phase 1.
- **Do we have user conviction?** Not yet validated. *Plan:* Run 8–10 moderated interviews and a prototype test with 30 applicants to test H1–H3 before committing to Phase 1.

7. Positioning

Use case	Pain point	Possible solution	Impact of solution (hypothetical)
Check application status	Ambiguous or stale progress	Verified status panel with timestamp	Status-related contacts down 20%

Use case	Pain point	Possible solution	Impact of solution (hypothetical)
Understand a requirement	Generic or buried help text	Contextual, approved explanations	Field drop-off down 10–15%
Recover from an error	Repeat errors; restarts	Taxonomy-based recovery	Error-related abandonment down 15%
Reach a human	Retelling; lost context	Structured handoff summary	AHT down 25%; repeat explanations down 40%
Understand data requests	Distrust; unclear purpose	Plain-language data explanations	Data-request abandonment down 8%

8. Measuring Success

Product metrics

- **Self-serve resolution rate:** Sessions ending with a completed next action, no handoff, and no repeat contact within 7 days.
- **Application completion rate:** Completions divided by starts, by cohort.
- **Status-related contact rate:** Contacts per 100 applications, tracked against baseline.
- **Handoff rate and CSAT:** Share of sessions escalated; post-handoff satisfaction.
- **Retention:** Return use within 30 days among applicants with stalled applications.

AI-specific metrics

- **Grounding accuracy:** Status claims matching the system of record. *Target:* 99.5% or higher. *Critical threshold:* any confirmed incorrect status triggers review.
- **Hallucinated-policy rate:** Responses citing content not in the approved store. *Target:* 0.
- **Classification accuracy (taxonomy):** Precision 90% or higher, recall 85% or higher, on the gold set.
- **Appropriate-escalation rate:** Handoffs triggered when the assistant should not answer. *Target:* 95% or more of low-confidence cases escalated.
- **Sales-content leakage:** *Target:* 0 in production sample audits.
- **Latency:** p95 under 2.5 seconds.

What quality means strategically: Customers can trust what the assistant says, and it knows when to stop. A confidently wrong answer is a launch blocker. Launch when grounding and leakage targets are met on the gold set and a 2-week shadow run is complete.

North Star metric

Verified self-serve resolutions per 1,000 active applicants per week: Sessions where a customer completes a verified next action without handoff and without repeat contact within 7 days. This captures user value, trust, and business efficiency together.

9. Launching

Stakeholders and communication

- **Compliance and Legal:** Content approval, disclosure review, data-processing terms. Gate before each phase.
- **Engineering and Data:** Integration, telemetry, and evaluation pipeline. Weekly sync.
- **Support Operations:** Handoff design, agent training, and feedback loop. Biweekly review.
- **Sales:** Informed of the separation policy; no assistant-driven sales pathways.
- **Customers:** In-product disclosure that they are using an AI assistant; clear path to a human; updated privacy notice.
- **Executive sponsor:** Monthly dashboard of North Star, AI metrics, and risk register.

Roll-out strategy

1. **Internal dogfood (2 weeks):** Staff use the assistant on test applications.
2. **Shadow mode (2 weeks):** Assistant generates responses but does not show them; compare against agent answers.
3. **Pilot (4 weeks):** 5% of eligible applicants, with kill switch and daily monitoring.
4. **Expansion:** 25% → 50% → 100%, gated on North Star, grounding accuracy, and complaint thresholds.
5. **Ongoing:** Monthly taxonomy review; quarterly content audit; annual privacy and model risk review.

Go-to-market considerations: Position as a service feature, not a sales channel. Communicate transparently with customers. Coordinate with Support to avoid a sudden drop in phone capacity during the pilot, and keep static help available as a fallback.

Immediate next steps

- [ ] Confirm product name, customer segment, and regulatory scope.
- [ ] Validate H1–H3 with 8–10 interviews and a 30-user prototype test.
- [ ] Obtain Compliance sign-off on the approved content model.
- [ ] Sample historical contact data to build the error taxonomy and gold set.
- [ ] Produce budget estimate for Phase 0–1.